Project Insights

1. Missing Data Patterns

Initial analysis of the dataset showed that several features contained missing values, especially age, BMI, cholesterol, glucose, gender, and region. Approximately 8–12% of the records contained missing entries, which could negatively impact machine learning model performance if left untreated.

To address this issue, multiple imputation strategies were applied:

- Numerical features such as age, BMI, cholesterol, and glucose were filled using mean imputation.
- Categorical features such as gender and region were filled using most frequent value imputation.
- Advanced techniques including KNN Imputer and MICE (Iterative Imputer) were applied to estimate missing values based on relationships between features.

These techniques ensured that all missing values were handled without removing useful data records, preserving the dataset size.

2. Outlier Detection and Treatment

During exploratory data analysis, several extreme values were identified in medical attributes such as BMI, cholesterol, blood pressure, and glucose levels. These outliers were likely caused by measurement errors or abnormal spikes in the synthetic dataset.

Different statistical methods were used to detect and treat these outliers:

- Z-Score Method was used to identify extreme values in cholesterol and glucose levels.
- IQR Method was applied to detect abnormal BMI values.
- Percentile Method removed extreme values outside the 1st and 99th percentile range.
- Winsorization Technique capped extreme cholesterol values instead of removing them completely.

Using multiple techniques helped reduce noise in the dataset while preserving most of the useful information.

3. Dataset Comparison (Before vs After Cleaning)

Metric	Before Preprocessing	After Preprocessing
Missing Values	Present in multiple columns	0
Outliers	High in BMI, cholesterol, glucose	Reduced
Dataset Quality	Inconsistent and noisy	Clean and structured
Data Usability	Limited for ML models	Suitable for ML

After preprocessing, the dataset became more stable and statistically reliable.

4. Data Quality Improvement

Before preprocessing:

- Dataset contained missing values and abnormal spikes.
- Feature distributions were skewed due to extreme outliers.

- Data inconsistencies could reduce model accuracy.

After preprocessing:

- Missing values were successfully handled.
- Extreme values were either removed or capped.
- Feature distributions became more balanced and realistic.

This significantly improved data quality and reliability for analysis.

5. Dataset Usability for Machine Learning

The cleaned dataset is now suitable for machine learning tasks, including:

- Heart disease risk prediction
- Patient health classification
- Healthcare analytics and risk modeling

The target variable disease_risk (0 = Low Risk, 1 = High Risk) was kept unchanged to ensure proper supervised learning.

6. Final Dataset Summary

After completing the preprocessing steps:

- All missing values were handled
- Outliers were significantly reduced
- Feature distributions became stable

The final cleaned dataset was exported as:

final_clean_dataset.csv

This dataset is now machine learning ready and suitable for predictive modeling.

7. Key Learning Outcomes

Through this project, the following skills were developed:

- Performing missing value analysis
- Applying multiple imputation techniques
- Detecting outliers using Z-Score, IQR, and Percentile methods
- Applying Winsorization for outlier treatment
- Improving dataset quality before machine learning

This project highlights that data preprocessing is one of the most critical steps in the data science pipeline, as model performance depends heavily on clean and reliable data.